

Mathematics Worksheet - Class 10

Topics Covered: Quadratic Equations, Algebra

Total Questions: Multiple sections below

Multiple Choice Questions (MCQs)

1. If the quadratic equation ($x^2 - 5x + 6 = 0$) is solved, what are the roots?
 - a) 2 and 3
 - b) -2 and -3
 - c) 1 and 6
 - d) -1 and -6
 2. The sum of the roots of the quadratic equation ($2x^2 - 7x + 3 = 0$) is:
 - a) $(\frac{7}{2})$
 - b) $(-\frac{7}{2})$
 - c) $(\frac{3}{2})$
 - d) $(-\frac{3}{2})$
 3. Which of the following is a factor of the quadratic expression ($x^2 + 7x + 12$)?
 - a) $(x + 3)$
 - b) $(x - 3)$
 - c) $(x + 4)$
 - d) Both a and c
 4. If (p) and (q) are roots of the quadratic equation ($x^2 - 4x + 1 = 0$), then the value of ($p^2 + q^2$) is:
 - a) 14
 - b) 18
 - c) 16
 - d) 12
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Short Answer Questions

1. Find the roots of the quadratic equation ($3x^2 - 2x - 1 = 0$) using the quadratic formula.
2. Express the quadratic polynomial ($x^2 - 9x + 20$) as a product of two linear factors.
3. If one root of the quadratic equation ($x^2 - 6x + k = 0$) is 2, find the value of (k).
4. Simplify and factorize the algebraic expression ($4x^2 - 25$).
5. Write the quadratic equation whose roots are 5 and (-3).

Long Answer Questions

1. A rectangular garden has an area of $(48, \text{m}^2)$. The length of the garden is 4 meters more than its breadth. Form a quadratic equation to represent this situation and find the dimensions of the garden.
 2. The product of two consecutive positive integers is 72. Form a quadratic equation to represent this and find the integers.
 3. Solve the quadratic equation $(2x^2 + 3x - 5 = 0)$ by completing the square method.
 4. The sum of the squares of two consecutive integers is 365. Find the integers by forming and solving a quadratic equation.
 5. A ball is thrown upward with a velocity of 20 m/s from the top of a building 45 m high. The height (h) (in meters) of the ball after (t) seconds is given by $(h = -5t^2 + 20t + 45)$. After how many seconds will the ball hit the ground?
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Word Problems

1. The difference between the squares of two numbers is 180. If one number is 15, find the other number.
 2. A shopkeeper sells a certain number of pens at ₹5 each and loses ₹20. If he sells 4 more pens at ₹5 each, he gains ₹10. Find the number of pens he originally intended to sell.
 3. The length of a rectangle is 3 meters more than twice its breadth. If the area is $135 (\text{m}^2)$, find the dimensions of the rectangle.
 4. The sum of the reciprocals of two numbers is $(\frac{5}{6})$, and the sum of the numbers is 10. Find the numbers.
 5. A train travels a certain distance at a speed of 60 km/h and returns at a speed of 40 km/h. If the total time for the journey is 5 hours, find the distance between the two stations.
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Answer Key

Multiple Choice Questions

1. a) 2 and 3
2. a) $(\frac{7}{2})$
3. d) Both a and c
4. a) 14

Short Answer Questions

1. Roots: $(x = 1), (x = -\frac{1}{3})$
2. $((x - 4)(x - 5))$
3. $(k = 8)$
4. $((2x - 5)(2x + 5))$

5. $(x^2 - 2x - 15 = 0)$

Long Answer Questions

1. Dimensions: Breadth = 4 m, Length = 8 m
2. Integers: 8 and 9
3. Roots: $(x = 1)$, $(x = -\frac{5}{2})$
4. Integers: 12 and 13
5. Time to hit ground: 5 seconds

Word Problems

1. Other number = 27
2. Number of pens = 40
3. Breadth = 9 m, Length = 21 m
4. Numbers = 4 and 6
5. Distance = 120 km